

Discussions

Acidity-mediated space charge potential at heterointerfaces and GBs

Based on the relative acidity of infiltrants or substrates relative to PCO, adjacent acidic/basic species were expected to induce electron depletion/accumulation at the heterointerfaces with PCO_{NA}, thereby leading to increased/decreased resistance along both the conduction pathways through substrate (R_2) and surface (R_5) heterointerfaces. As we have shown in **Fig. 2c** and **3b**, elements with higher acidity relative to PCO (Al_2O_3 and SiO_2) resulted in decreases in conductivity, while elements with lower acidity relative to PCO (MgO and Li_2O) resulted in increases in conductivity, consistent with our expectation. However, several details need to be explained in more detail to fully appreciate the observed trends:

I) The relative magnitude of change in the conductivity of PCO_{NA} driven by surface acidity modulation with basic and acidic elements compared to uninfiltrated samples differed significantly as illustrated in **Fig. 2c** and **3b**. This divergence might originate from the distinct conduction pathways determined by space charge potentials at the heterointerfaces. For example, in the case of acidic treatments, the higher relative acidity compared to that of PCO is expected to generate a more resistive conduction pathway through the surface/substrate heterointerfaces (R_5/R_2), leading to the expectation that conduction occurs primarily through the bulk (R_1) (**Fig. 4a** and **4c**). On the other hand, the treatment with basic elements will result in carrier enhancement due to electron accumulation and thereby enhanced conduction through the surface/substrate heterointerface (R_5/R_2) (**Fig. 4b**). Interestingly, Li-infiltration had a much larger impact (three orders of magnitude at 450 °C) on the PCO_{NA} conductivity than the MgO substrate (factor of ~ 10 at 450 °C), although both have a higher basicity than PCO. This can be related to (i) the very different natures of the two interfaces (the unusual droplet-shape of the top rough surface of PCO_{NA} leads to a much higher interface area than at the planar substrate interface) and (ii) the higher relative basicity of Li_2O is expected to result in higher electron accumulation and therefore higher conductivity compared to MgO.

II) In the case of acidic treatment with SiO_2 infiltration or use of Al_2O_3 substrate, one would expect, as mentioned above, the conductivity to solely conduct through the bulk (R_1), and therefore for it to reach similar values to that of the bulk PCO20 reference. However, the conductivity and its E_a for PCO_{NA} are reduced in both cases compared to bulk PCO20 (1 eV), which indicates that conduction is impacted by pathways in addition to the bulk conduction (R_1). Here one must consider how grain boundaries (GBs), as well, can potentially serve as

either short-circuiting pathways (R₃) or serve to block transport (R₄), as illustrated in **Fig. 1a** and **1b**. While we do not report any direct measurements of GB characteristics in this study, based on their expected behavior, we can explain the above observations. In the following, we therefore briefly review key relevant findings describing the roles of GBs in impacting charge transport in ceria-based systems and identify how small polaron hopping mobility, characteristic of electron transport in PCO, can result in GB phenomena contrary to normal expectations. For those interested, more detailed discussions of these issues may be found in **Supplementary Note 2**.

GBs in ceria systems^{37,38}, in general, exhibit a positive space charge potential associated with a positive GB core charge, causing a depletion of oxygen vacancies and accumulation of electrons in the adjacent space charge zones³⁹. In nominally undoped CeO₂, the accumulation of electrons has been shown to lead to electronically conducting layers that bypass bulk ionic transport, thereby controlling the overall conductivity^{40–42}. On the other hand, as mentioned above, this same positive GB space charge potential in acceptor-doped CeO₂ (e.g. Gd doped CeO₂ - GDC), an oxygen ion conductor, can lead to orders of magnitude reduction in oxygen ion conduction due to oxygen vacancy depletion at the GBs^{37,43}. As recently demonstrated by the authors, this space charge potential can be tuned by in-diffusion of substrate elements (i.e. Mg and Al) into GDC thin films at intermediate temperatures (700-900 °C)⁴³, as described in **Supplementary Note 2**. We suspect correspondingly significant Al up-diffusion occurred along GBs in PCO_{NA} samples grown on alumina substrates annealed at 700 °C (**Fig. 4c, lower image**), compared to those annealed at 400 °C (**Fig. 4a**), leading to an increase of the positive space charge potential at the GBs (**Fig. 4c**). While conduction along the GBs (R₃), in principle, should give enhanced conduction due to accumulation of electrons around GBs, that transport path nevertheless would have to pass through a mobility blockage related to the small polaron hopping controlling electronic transport in PCO connected with the path perpendicular to GBs (R₄), as discussed in **Supplementary Note 2** and displayed in **Supplementary Fig. 6** and **7**. This results in reduced E_a , attributed to the enhanced space charge potential at the GBs, yet without a corresponding increase in conductivity. In fact, it is even lower than that of the PCO_{NA} annealed at 400 °C (**Supplementary Fig. 8**), where Al up-diffusion is expected to be kinetically restricted and therefore the space charge potentials at the grain boundaries are lower. The lower conductivity is related to the inability of electrons to hop from occupied Pr³⁺ sites to adjacent unoccupied Pr⁴⁺ sites under conditions of high electron accumulation at the GBs. XRD spectra showed no difference in peak position and shape of PCO_{NA} annealed at different

temperatures suggesting that no differences in strain were present (**Supplementary Fig. 9**). On the other hand, in the case of the nanowires sitting on MgO, transport across GBs (R₄) become less resistive due to the up-diffusion of Mg at the GBs, lowering the space charge potential, allowing for the measurement of the un-impeded conductivity enhancement parallel to the substrate heterointerface (R₂) (**Fig. 4b**). Thus, the similarities in E_a between the PCO_{NA} treated with acids and bases, but different conductivity trends, originate from the presence of positive space charge potential at different dominant conduction pathways.

Interestingly, opposite trends were observed in a purely ionic conducting ceria system, 3 mol% Gd-doped ceria (GDC3), prepared via the same method as PCO_{NA} (**Supplementary Note 3**). Here the basic treatment with MgO as a substrate leads to a total decrease in conductivity in contrast to Al₂O₃. This observation aligns very well with the space charge model, as the majority of carriers in this case are positively charged oxygen vacancies ($V_o^{\bullet}$), and therefore the space charges present at the heterointerface are expected to impact the carrier in the opposite direction to the electronic carriers in PCO_{NA}. The activation energies measured in the GDC3 nanowire arrays are characteristic of grain boundary-controlled resistance and are clearly influenced by the substrate heterointerfaces, similar to the case of the PCO_{NA} (**Supplementary Fig. 10**). Therefore, the Smith acidity scale is a universally applicable descriptor for predicting the influence of heterointerface space charge effects not only on electronic but also ionic carriers.

III) Finally, the relatively moderate enhancement achieved in sequential infiltration with Li-species following Si (**Fig 2c**), compared to solely Li-infiltrated PCO_{NA}, can be attributed to the effective Smith acidity of Li-Si compounds⁴⁴. As evidenced by the XPS spectra (**Supplementary Fig. 11**), Li-species react with SiO₂ at high temperatures, forming intermediate lithium silicate composites (Li₂SiO₃ or Li₄SiO₄). The acidity of these composites should be located between the Smith acidity scale of Li₂O and SiO₂, which can be expressed as:

$$a_{\text{Li}_2\text{SiO}_3} = \frac{2a_{\text{Li}_2\text{O}} + a_{\text{SiO}_2}}{3} \text{ or } a_{\text{Li}_4\text{SiO}_4} = \frac{4a_{\text{Li}_2\text{O}} + a_{\text{SiO}_2}}{5}$$

Due to the higher basicity of Li₂O and its dominant influence, lithium silicate exhibits strong basicity, in contrast to SiO₂, and plays a role in accumulating electrons at the top heterointerface of PCO_{NA}, although obviously not as high as Li₂O alone. This leads to the formation of a dominant conduction pathway through the surface heterointerface (R₅). This change in effective Smith acidity leads to a 100-fold improvement in conductivity.